

SIH26038 — Complete Project Blueprint

Explainable AI for Diabetic Retinopathy Screening in Rural India

Systematic interpretation, technical requirements, recommended implementation decisions, architecture, workflow, evaluation and final checklist.

1. PS in One Simple Paragraph

The project asks for a MATLAB-based AI system that takes a retinal/fundus photograph, checks whether the image is good enough, improves poor-but-usable images, rejects unusable images, identifies important retinal structures and lesions, grades diabetic retinopathy from Level 0 to Level 4, and explains why the AI gave that grade. The system should also simulate the complete rural telemedicine workflow in Simulink, considering image acquisition, bandwidth, processing speed and ophthalmologist review capacity for programs handling 100,000+ patients/year. The final prototype must achieve more than 90% sensitivity and more than 85% specificity for referable DR (Level 2+), provide clinically useful Grad-CAM explanations, and demonstrate that the integrated approach performs better than individual techniques.

2. The Big Picture

```
RETINAL IMAGE
↓
IMAGE QUALITY CHECK
↓
Good → Continue
Borderline → Enhance → Recheck
Bad → Reject → Recapture
↓
RETINAL ANALYSIS
Disc / Fovea / Vessels / Lesions
↓
DR SEVERITY GRADING
Level 0 → 4
↓
EXPLAINABLE AI
Grad-CAM + lesions + confidence + report
↓
SIMULINK SIMULATION
Rural telemedicine workflow
```

3. Image Quality Assessment

Required: automatic evaluation of focus, illumination and field of view; adaptive enhancement using CLAHE, illumination normalization and denoising; rejection of bad/ungradeable images with recapture feedback.

- Focus/blur: a sharpness measure such as variance of Laplacian.
- Illumination: assess too dark, too bright or uneven lighting.
- Field of view: check whether sufficient retina is visible.
- Combine quality checks into GOOD, ENHANCE or UNGRADEABLE.
- Choose thresholds from validation data rather than arbitrary medical thresholds.

4. Image Enhancement

- CLAHE improves local contrast.
- Illumination normalization reduces uneven lighting.
- Denoising removes unwanted noise while preserving useful structures.
- For borderline images: enhance → recheck → continue or request recapture.

5. Retinal Structure and Lesion Analysis

Required: optic disc localization, fovea localization, vessel segmentation, microaneurysm detection, exudate segmentation, hemorrhage classification and neovascularization detection.

- Optic disc: bright circular region where the optic nerve exits the eye.
- Fovea: central region responsible for sharp vision.
- Blood vessels: retinal vessel network.
- Microaneurysms: tiny abnormal bulges in retinal blood vessels.
- Exudates: bright/yellowish deposits associated with retinal damage.
- Hemorrhages: areas of blood leakage in retinal tissue.
- Neovascularization: abnormal growth of new blood vessels.

6. Dataset Plan

The PS names APTOS 2019, IDRiD, DRIVE and Messidor-2. Use them for different tasks rather than simply mixing them.

Dataset	Primary role
APTOS 2019	DR severity classification
IDRiD	DR grading + lesion information
DRIVE	Blood-vessel segmentation
Messidor-2	External validation / DR grading

7. Why IDRiD Is Especially Important

IDRiD is particularly useful for lesion analysis because the project needs clinically meaningful lesion evidence. It can support DR grading and lesion-level analysis such as microaneurysms, hemorrhages and exudates.

8. DRIVE — Why We Need It

DRIVE is appropriate for the vessel-extraction requirement. The conceptual output is a vessel map produced from the retinal image using a segmentation model.

9. DR Severity Grading

Required five levels:

- Level 0 — No DR
- Level 1 — Mild NPDR
- Level 2 — Moderate NPDR
- Level 3 — Severe NPDR
- Level 4 — Proliferative DR

Referable DR: Level 2+ is referable. Therefore produce both a five-class prediction (0–4) and a screening decision: Levels 0–1 non-referable; Levels 2–4 referable.

10. CNN Architecture

Do not flatten the RGB image before convolution. CNN needs the spatial arrangement of pixels.

```
RGB image: H × W × 3
↓
Convolution
↓
Feature maps
↓
More convolution
↓
More feature maps
```

↓
Global Average Pooling
↓
1D feature vector
↓
Classifier
↓
DR Level 0-4

Your current workflow uses EfficientNet-B3. This is a suitable primary backbone for DR classification unless MATLAB integration requires a change. EfficientNet-B3 performs convolutional feature extraction; Global Average Pooling can then produce a compact 1D feature vector.

Simple CNN for learning

Input RGB Image → Conv2D → ReLU → MaxPooling
→ Conv2D → ReLU → MaxPooling → Conv2D → ReLU
→ GlobalAveragePooling2D → Dense → 5-class output

Flatten is possible, but for large feature maps Global Average Pooling is generally preferable because it produces a much smaller representation.

11. Important MATLAB Requirement

The PS calls for a MATLAB-based retinal image analysis pipeline and lists Image Processing Toolbox, Computer Vision Toolbox, Deep Learning Toolbox, Medical Imaging Toolbox, Simulink and Statistics and Machine Learning Toolbox.

Your current experiments use Google Colab, Python, PyTorch, CUDA and EfficientNet-B3. This is useful for rapid experimentation, but if the PS requirement is strict, the final demonstrable solution should be compatible with MATLAB/Simulink. A trained model may be imported into MATLAB where supported rather than necessarily retraining everything from scratch.

12. Explainable AI

- Grad-CAM shows where the AI looked when making its decision.
- Lesion-level evidence connects the prediction to clinically meaningful regions.
- Calibrated confidence makes reported probability more meaningful.
- Automated report combines quality, grade, referability, confidence and evidence.

Example report
Image Quality: GOOD
DR Grade: Level 3
Referable DR: YES
Confidence: 91%
Evidence: Microaneurysms, Hemorrhages, Exudates
AI Attention: Grad-CAM
Recommendation: Ophthalmologist review

The output should enable ophthalmologist validation in under 30 seconds.

13. Simulink

Model the rural telemedicine workflow including image acquisition rate, bandwidth constraints, processing throughput and review capacity for district-level programs handling 100,000+ patients annually.

Patients → Fundus Camera → Image Acquisition → Network/Bandwidth
→ AI Processing → Automatic Screening → High-risk Cases
→ Ophthalmologist → Final Review

The simulation should identify throughput, bottlenecks, waiting time and the effect of AI/human-review capacity.

14. 100,000+ Patients/Year

100,000 patients/year is approximately 274 patients/day averaged over 365 days. The Simulink model should use realistic assumptions for arrival rate, capture, transmission, AI processing and human review rather than simply claiming scalability.

15. Performance Requirements

Metric	Required / target
Referable DR sensitivity	>90%
Referable DR specificity	>85%
DR classes	5 (0–4)
Explainability	Grad-CAM + clinical evidence
Confidence	Calibrated
Human review target	<30 seconds
Simulation	Simulink
Scale	100,000+ patients/year

16. Additional Evaluation Metrics

- Accuracy
- Precision
- Recall / sensitivity
- Specificity
- F1-score
- Confusion matrix
- Quadratic Weighted Kappa (QWK)
- TP, TN, FP, FN for referable DR

17. Class Imbalance

DR datasets may contain many more images in some severity classes. Recommended approaches include class weights, a weighted sampler, or carefully designed augmentation; compare strategies on validation data.

18. Data Splitting and Leakage

Use separate training, validation and test data. A reasonable starting point is 70/15/15, although the PS does not mandate a ratio. More important is strict prevention of leakage: the same patient/image must not appear across training and testing. Keep external validation data independent where possible.

19. Data Augmentation

The PS does not specify augmentation. Recommended medically sensible transformations include modest rotation, horizontal flip where appropriate, small crop/resize changes, and limited brightness/contrast variation. Avoid aggressive transformations that create unrealistic retinal anatomy.

20. Microaneurysm Detection

The PS calls for sub-pixel microaneurysm detection. Tiny lesions need sufficient resolution. For a prototype, investigate a dedicated high-resolution detection/segmentation approach rather than relying only on the global DR classifier.

21. Complete Integrated Architecture

```
RGB FUNDUS IMAGE
↓
IMAGE QUALITY ASSESSMENT
↓
GOOD / BORDERLINE / UNGRADEABLE
↓
ENHANCEMENT (if needed)
```

↓
 RETINAL ANALYSIS
 ■■ Vessel segmentation
 ■■ Lesion detection
 ■■ Disc / fovea localization
 ↓
 FEATURE EXTRACTION
 ↓
 CNN / EfficientNet
 ↓
 5-CLASS DR GRADING
 ↓
 LEVEL 0–4 + REFERABLE?
 ↓
 EXPLAINABLE AI
 ■■ Grad-CAM
 ■■ Lesion evidence
 ■■ Calibrated confidence
 ↓
 AUTOMATED REPORT
 ↓
 HUMAN REVIEW
 ↓
 SIMULINK TELEMEDICINE MODEL

22. Explicit Requirements vs Our Decisions

Explicitly required by the PS

- MATLAB-based pipeline
- Image quality: focus, illumination, field of view
- CLAHE, illumination normalization, denoising
- Reject ungradeable images and recapture feedback
- Optic disc/fovea localization
- Vessel segmentation
- Microaneurysm detection
- Exudate segmentation
- Hemorrhage classification
- Neovascularization detection
- DR Levels 0–4 and Level 2+ referable DR
- >90% sensitivity and >85% specificity
- Grad-CAM, lesion-level evidence, calibrated confidence
- Automated report and <30-second review target
- Simulink acquisition/bandwidth/throughput/review modelling
- 100,000+ patients/year
- Clinical validation and published benchmark comparison
- Integrated system outperforming single techniques

Not specified exactly — recommended decisions

- Primary classifier: EfficientNet-B3, subject to MATLAB compatibility.
- Start classification experiments at 224×224×3; preserve higher resolution for tiny-lesion analysis.
- Strict train/validation/test separation and leakage prevention.
- Moderate medically sensible augmentation.

- Class weighting or other balancing after validation comparison.
- Set quality thresholds using validation data.
- Use temperature scaling or another validated calibration method.
- Report accuracy, precision, recall, specificity, F1, confusion matrix and QWK.
- Use external validation and baseline comparisons to strengthen claims.

23. Baseline vs Integrated System

Baseline 1: Raw image → CNN → DR classification
 Baseline 2: Enhanced image → CNN → classification

Proposed:
 Quality assessment + Enhancement + Retinal structures
 + Lesion information + CNN grading + Explainability

Compare all under the same evaluation protocol.

This provides evidence that the integrated pipeline adds value rather than simply containing more components.

24. Clinical Validation

The PS asks for clinical validation rigor. For a student prototype, avoid overstating real-world clinical validation. Stronger evidence can come from independent test data, published benchmarks, clinical criteria and expert-oriented explanations. If an ophthalmologist is available, ask them to evaluate explanation usefulness, highlighted lesion correctness, report readability and confidence usefulness.

25. Final Demo

SCREEN 1 – Quality
 Image Quality: GOOD | Focus: Good | Illumination: Good | FOV: Good

SCREEN 2 – Analysis
 Optic Disc: Detected | Fovea: Detected | Vessels: Segmented
 Lesions: Detected

SCREEN 3 – DR
 Grade: Level 3 | Referable: YES | Confidence: 92%

SCREEN 4 – Explainability
 Original image + Grad-CAM + lesion annotations

SCREEN 5 – Report
 Automated DR Screening Report

SCREEN 6 – Simulink
 Patients/year + throughput + bandwidth + waiting time
 + ophthalmologist capacity

26. Technology Stack

MATLAB
 ■■ Image Processing Toolbox
 ■■ Computer Vision Toolbox
 ■■ Deep Learning Toolbox
 ■■ Medical Imaging Toolbox
 ■■ Statistics and Machine Learning Toolbox
 ■■ Simulink

Python / Google Colab
 ■■ Rapid experimentation and model prototyping
 (if retained during development)

27. Dataset-to-Task Mapping

APTOS 2019 → 5-class DR grading / main classifier training
 IDRiD → DR grading + lesion analysis + explainability evidence
 DRIVE → Vessel segmentation

28. Project Build Order

- Phase 1 — Dataset: organize, understand labels, check formats, handle corrupted data.
- Phase 2 — Preprocessing: RGB handling, resize, normalization, quality assessment, enhancement.
- Phase 3 — Basic DR classifier: CNN / EfficientNet → 5-class classification.
- Phase 4 — Evaluation: confusion matrix, QWK, sensitivity, specificity, F1.
- Phase 5 — Retinal structures: vessels, optic disc and fovea.
- Phase 6 — Lesions: microaneurysms, exudates, hemorrhages, neovascularization.
- Phase 7 — Explainability: Grad-CAM, lesion evidence, calibrated confidence, report.
- Phase 8 — Integration: connect quality, segmentation, classification and explainability.
- Phase 9 — Simulink: acquisition, bandwidth, AI processing, review and 100,000+ workload.
- Phase 10 — Final validation: benchmarks, external validation, baseline comparison and targets.

29. The PS in Five Questions

1. Is the retinal image usable?
2. What retinal structures / lesions are present?
3. What is the DR grade (0-4)?
4. Why did the AI make that decision?
5. Can the rural telemedicine system handle the workload?

30. Final Project Checklist

- APTOS 2019
- IDRiD
- DRIVE
- Messidor-2
- RGB preprocessing
- Image quality
- Focus
- Illumination
- FOV
- CLAHE
- Illumination normalization
- Denoising
- Ungradeable rejection
- Recapture feedback
- Optic disc
- Fovea
- Vessel segmentation
- Microaneurysm detection
- Exudate segmentation
- Hemorrhage classification

- Neovascularization detection
- 5-class grading
- Level 2+ referable DR
- >90% sensitivity
- >85% specificity
- Grad-CAM
- Lesion evidence
- Calibrated confidence
- Automated report
- <30-second review
- MATLAB
- Simulink
- Acquisition modelling
- Bandwidth modelling
- Processing throughput
- Review capacity
- 100,000+ patients/year
- Published benchmark comparison
- Baseline vs integrated comparison
- External validation
- Final demo

31. Final Strategic Point

Do not implement all components simultaneously. Build progressively and check every stage against the PS. The central idea is: determine whether the image is usable, analyze retinal structures and lesions, grade DR from Level 0–4, explain the prediction, generate a concise report, and demonstrate that the complete workflow can scale in a rural telemedicine setting.

Source basis: This document is based on the SIH26038 problem statement supplied in the conversation; recommendations are explicitly presented as implementation decisions where the PS does not specify an exact method.